UNIT
9

ELECTRO CHEMISTRY



Walther Hermann Nernst

Walther Hermann Nernst, was a German chemist known for his work in thermodynamics, physical chemistry, electrochemistry, and solid state physics. His formulation of the Nernst heat theorem helped to pave the way for the third law of thermodynamics, for which he won the 1920 Nobel Prize in Chemistry. He is also known for developing the Nernst equation in 1887. He also derived the Nernst equation for the electrical potential generated by unequal concentrations of an ion separated by a membrane that is permeable to the ion. His equation is widely used in cell physiology and neurobiology.

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Learning Objectives

After learning this unit, the students will be able to

- Recognise the conductivity of electrolylic solution
- Define the terms resistivity, conductivity equivalent and molar conductivity
- Explain the variation of conductivity with concentration
- Apply Kohlrausch law to calculate the conductivity of weak electrolyte at infinite dilution.
- Describe an electrochemical cell
- Differentiate between an electrochemical and electolylic cell
- Represent a galvanic cell using IUPAC cell notation
- Derive Nernst equation and apply it to calculate E_{cell}.
- Define Faraday's Law of electrolysis
- Describe the construction of batteries
- Explain corrosion as an electrochemical process.

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